Stochastic Mathematics for Finance

Week 1 Resources

Probability, Moments, Markov Chains, and Random Walks

Purpose of Week 1. The aim of this week is to build the basic mathematical language needed for stochastic modelling in finance. Later topics such as the Central Limit Theorem, Brownian motion, stochastic calculus, stochastic differential equations, Black–Scholes, Monte Carlo simulation, and jump processes all rely on the ideas introduced here.

The main conceptual goal is to understand how randomness evolves over time.

By the end of this week, you should be comfortable with:

- probability spaces, events, conditional probability, and independence;
- random variables and common probability distributions;
- expectation, variance, covariance, correlation, and moments;
- stochastic processes as collections of random variables indexed by time;
- Markov chains and transition probability matrices;
- communicating classes, irreducibility, recurrence, transience, periodicity, and other class properties;
- stationary distributions;
- random walks and their connection to Brownian motion and finance;
- basic simulation of stochastic processes in Python.

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1 Why Probability and Stochastic Processes Matter in Finance

Finance is full of uncertainty. Stock prices, interest rates, exchange rates, option payoffs, credit defaults, and market crashes cannot be predicted exactly.

Instead of trying to predict the future perfectly, stochastic modelling gives us a way to describe uncertainty mathematically.

Some examples:

- The return of a stock tomorrow can be modelled as a random variable.
- The price of a stock over time can be modelled as a stochastic process.
- Market regimes such as bull, neutral, and bear markets can be modelled using Markov chains.
- Random walks provide a basic model for price movements.
- Brownian motion, which is central to Black–Scholes theory, can be understood as a scaling limit of random walks.
- Monte Carlo simulation uses repeated random sampling to estimate prices and risks.

The goal of this project is not only to learn formulas, but also to understand how stochastic models are built, analyzed, and simulated.

2 Basic Probability

2.1 Random Experiments

A **random experiment** is an experiment whose outcome is uncertain before it is performed.

Examples:

- Tossing a coin.
- Rolling a die.
- Observing tomorrow's stock return.
- Checking whether a borrower defaults.
- Observing whether a market moves up or down tomorrow.

2.2 Sample Space

The **sample space** is the set of all possible outcomes of a random experiment. It is usually denoted by

$$\Omega.$$

Example 1. *If we toss a coin once, then*

$$\Omega = \{H, T\}.$$

If we roll a die once, then

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

2.3 Events

An **event** is a subset of the sample space.

Example 2. *If a die is rolled, then the event that the outcome is even is*

$$A = \{2, 4, 6\}.$$

If A is an event, then $P(A)$ denotes the probability that event A occurs.

2.4 Basic Rules of Probability

For any event A,

$$0 \leq P(A) \leq 1.$$

The probability of the whole sample space is

$$P(\Omega) = 1.$$

The probability of the impossible event is

$$P(\emptyset) = 0.$$

If A^c is the complement of A, then

$$P(A^c) = 1 - P(A).$$

For two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are disjoint, meaning

$$A \cap B = \emptyset,$$

then

$$P(A \cup B) = P(A) + P(B).$$

2.5 Conditional Probability

The conditional probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)},$$

provided

$$P(B) > 0.$$

This measures the probability of A after we know that B has occurred.

Example 3. *Suppose*

$$P(A) = 0.4, \qquad P(B) = 0.5, \qquad P(A \cap B) = 0.2.$$

Then

$$P(A \mid B) = \frac{0.2}{0.5} = 0.4.$$

2.6 Independence

Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Equivalently, if $P(B) > 0$, then

$$P(A \mid B) = P(A).$$

This means that knowing B occurred does not change the probability of A.

2.7 Why Dependence Matters in Finance

In finance, independence is often too strong an assumption. During normal market conditions, two assets may appear weakly related. But during crises, many assets can fall together.

For example, if two positions have loss events L_1 and L_2, then assuming independence gives

$$P(L_1 \cap L_2) = P(L_1)P(L_2).$$

But in reality, joint losses may be much more likely than this. This is why dependence and correlation are central in risk management.

3 Random Variables

3.1 Definition

A **random variable** is a function from the sample space to the real numbers.

Formally,

$$X : \Omega \to \mathbb{R}.$$

A random variable assigns a numerical value to each outcome of a random experiment.

Example 4. *If a coin is tossed, define*

$$X = \begin{cases} 1, & \text{if the outcome is heads,} \\ 0, & \text{if the outcome is tails.} \end{cases}$$

Then X is a random variable.

3.2 Discrete Random Variables

A random variable is **discrete** if it takes countably many possible values.

Examples:

- Bernoulli random variable.
- Binomial random variable.
- Poisson random variable.
- Number of defaults in a portfolio.
- Number of upward moves in a stock over n days.

3.3 Probability Mass Function

For a discrete random variable X, the probability mass function, or PMF, is

$$p_X(x) = P(X = x).$$

It satisfies

$$p_X(x) \geq 0$$

and

$$\sum_x p_X(x) = 1.$$

3.4 Continuous Random Variables

A random variable is **continuous** if it can take values in an interval or continuum of values.

Examples:

- A normally distributed return.
- Waiting time until an event.
- A continuously modelled interest rate.

3.5 Probability Density Function

For a continuous random variable X, probabilities are described using a probability density function, or PDF, $f_X(x)$.

The density satisfies

$$f_X(x) \geq 0$$

and

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1.$$

For an interval $[a, b]$,

$$P(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

For a continuous random variable,

$$P(X = x) = 0$$

for any single point x.

3.6 Cumulative Distribution Function

The cumulative distribution function, or CDF, of a random variable X is

$$F_X(x) = P(X \leq x).$$

The CDF exists for both discrete and continuous random variables.

4 Common Probability Distributions

4.1 Bernoulli Distribution

A Bernoulli random variable models a single success/failure experiment.

$$X \sim \text{Bernoulli}(p)$$

means

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p.$$

Examples:

- A stock goes up tomorrow or not.
- A borrower defaults or not.
- A trade is profitable or not.

4.2 Binomial Distribution

A binomial random variable counts the number of successes in n independent Bernoulli trials.

$$X \sim \text{Binomial}(n, p)$$

means

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Example: number of up days in n trading days.

4.3 Geometric Distribution

A geometric random variable models the number of trials until the first success.

If

$$X \sim \text{Geometric}(p),$$

then

$$P(X = k) = (1 - p)^{k-1} p, \qquad k = 1, 2, \ldots.$$

Example: number of days until a stock first crosses a certain level.

4.4 Poisson Distribution

A Poisson random variable models the number of events occurring in a fixed interval.

If

$$X \sim \text{Poisson}(\lambda),$$

then

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

Examples:

- Number of trades in a short time interval.
- Number of defaults in a portfolio.
- Number of jumps in a jump process.

4.5 Uniform Distribution

A uniform random variable on $[a, b]$ has density

$$f_X(x) = \frac{1}{b - a}, \qquad a \le x \le b.$$

We write

$$X \sim \text{Uniform}(a, b).$$

All values in $[a, b]$ are equally likely in the density sense.

4.6 Exponential Distribution

An exponential random variable models waiting times.

If

$$X \sim \text{Exponential}(\lambda),$$

then

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

Example: waiting time until the next trade or default event.

4.7 Normal Distribution

A normal random variable has density

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

We write

$$X \sim \mathcal{N}(\mu, \sigma^2).$$

The normal distribution is important because of the Central Limit Theorem. However, financial returns often have heavier tails than a normal distribution.

5 Expectation, Variance, and Moments

So far, we have described random variables using their probability distributions. However, in many applications, especially in finance, we often want to summarize a random variable using a few important numerical quantities.

For example, suppose X represents the return of an asset tomorrow. We may want to know:

- What is the average or expected return?
- How much does the return fluctuate around its average?
- Is the distribution symmetric or skewed?
- Does the distribution have heavy tails?

These questions are answered using **moments**.

5.1 Expectation

The **expectation** of a random variable is its long-run average value.

If we repeat a random experiment many times and observe the values of a random variable X, then the expectation tells us the average value we expect to see in the long run.

5.1.1 Expectation of a Discrete Random Variable

Let X be a discrete random variable taking values

$$x_1, x_2, x_3, \dots$$

with probabilities

$$P(X = x_i) = p_i.$$

Then the expectation of X is defined as

$$\mathbb{E}[X] = \sum_i x_i P(X = x_i).$$

Equivalently,

$$\mathbb{E}[X] = \sum_i x_i p_i.$$

This is a weighted average of the possible values of X, where the weights are the probabilities.

5.1.2 Example: Expectation of a Die Roll

Let X be the outcome of a fair six-sided die. Then

$$P(X = k) = \frac{1}{6}, \qquad k = 1, 2, 3, 4, 5, 6.$$

Therefore,

$$\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}.$$

Thus,

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

Notice that 3.5 is not a possible die outcome. This is not a problem. The expectation is not necessarily a value that the random variable can actually take. It is the long-run average.

5.1.3 Example: Expectation of a Bernoulli Random Variable

Let

$$X \sim \text{Bernoulli}(p).$$

Then

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p.$$

Therefore,

$$\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p.$$

So the expectation of a Bernoulli random variable is simply the probability of success.

5.1.4 Expectation of a Continuous Random Variable

If X is a continuous random variable with probability density function $f_X(x)$, then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

This is the continuous analogue of the discrete weighted average.

5.1.5 Example: Uniform Distribution

Let

$$X \sim \text{Uniform}(0, 1).$$

Then

$$f_X(x) = 1, \qquad 0 \leq x \leq 1.$$

Therefore,

$$\mathbb{E}[X] = \int_0^1 x\, dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}.$$

So the expected value of a uniform random variable on $[0, 1]$ is the midpoint of the interval.

5.2 Linearity of Expectation

One of the most important properties of expectation is linearity.

For any random variables X and Y, and constants $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

More generally,

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i].$$

A very important point is that **linearity of expectation does not require independence**.

Even if $X_1, \ldots, X_n$ are dependent,

$$\mathbb{E}[X_1 + \cdots + X_n] = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n].$$

5.2.1 Example

Suppose $X_1, \ldots, X_n$ are iid Bernoulli random variables with parameter p, and

$$S_n = X_1 + \cdots + X_n.$$

Then S_n counts the number of successes in n trials.

Using linearity of expectation,

$$\mathbb{E}[S_n] = \mathbb{E}[X_1 + \cdots + X_n] = \sum_{i=1}^{n} \mathbb{E}[X_i] = np.$$

Thus, if $S_n \sim \text{Binomial}(n, p)$, then

$$\mathbb{E}[S_n] = np.$$

5.3 Variance

Expectation tells us the average value of a random variable, but it does not tell us how spread out the values are.

For example, two random variables may have the same expectation but very different levels of uncertainty.

The **variance** of a random variable measures how far the random variable typically is from its mean.

The variance of X is defined as

$$\text{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

If we write

$$\mu = \mathbb{E}[X],$$

then

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

The quantity $X - \mu$ is the deviation of X from its mean. Squaring makes all deviations nonnegative and gives more weight to large deviations.

5.3.1 Computational Formula for Variance

A very useful formula is

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

This is often easier to compute than using the definition directly.

5.3.2 Derivation

Let $\mu = \mathbb{E}[X]$. Then

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2].$$

Expanding the square,

$$(X-\mu)^2 = X^2 - 2\mu X + \mu^2.$$

Taking expectation,

$$\mathrm{Var}(X) = \mathbb{E}[X^2 - 2\mu X + \mu^2].$$

Using linearity of expectation,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2.$$

Since $\mu = \mathbb{E}[X]$,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - 2\mu^2 + \mu^2.$$

Therefore,

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.}$$

5.3.3 Example: Variance of a Bernoulli Random Variable

Let

$$X \sim \mathrm{Bernoulli}(p).$$

We know that

$$\mathbb{E}[X] = p.$$

Since X only takes values 0 and 1,

$$X^2 = X.$$

Therefore,

$$\mathbb{E}[X^2] = \mathbb{E}[X] = p.$$

Using the computational formula,

$$\mathrm{Var}(X) = p - p^2 = p(1-p).$$

So if

$$X \sim \mathrm{Bernoulli}(p),$$

then

$$\boxed{\mathrm{Var}(X) = p(1-p).}$$

5.3.4 Example: Variance of a Rademacher Random Variable

Let Y be defined by

$$Y = \begin{cases} 1, & \text{with probability } \frac{1}{2}, \\ -1, & \text{with probability } \frac{1}{2}. \end{cases}$$

Then

$$\mathbb{E}[Y] = 1 \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0.$$

Also,

$$Y^2 = 1$$

always. Therefore,

$$\mathbb{E}[Y^2] = 1.$$

Hence,

$$\mathrm{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = 1 - 0^2 = 1.$$

This random variable is important because it is used to define a symmetric random walk.

5.4 Standard Deviation

The variance is measured in squared units. To return to the original units of the random variable, we define the **standard deviation**.

The standard deviation of X is

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

For example, if X represents daily return, then $\mathrm{SD}(X)$ is a measure of the typical size of fluctuations in daily return.

In finance, standard deviation is often used as a measure of volatility.

5.5 Moments

Moments are quantities of the form

$$\mathbb{E}[X^k],$$

where k is a positive integer.

The k-th raw moment of X is defined as

$$m_k = \mathbb{E}[X^k].$$

Thus:

$$m_1 = \mathbb{E}[X]$$

is the first raw moment,

$$m_2 = \mathbb{E}[X^2]$$

is the second raw moment,

$$m_3 = \mathbb{E}[X^3]$$

is the third raw moment, and so on.

5.6 Raw Moments

The **k-th raw moment** of X is

$$\mathbb{E}[X^k].$$

Raw moments are taken around zero.

For a discrete random variable,

$$\mathbb{E}[X^k] = \sum_x x^k P(X = x).$$

For a continuous random variable with density f_X,

$$\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k f_X(x)\, dx.$$

5.6.1 Interpretation

The first raw moment is the mean:

$$\mathbb{E}[X].$$

The second raw moment,

$$\mathbb{E}[X^2],$$

is used to compute variance:

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Higher raw moments give more information about the shape of the distribution.

5.7 Central Moments

Raw moments are measured around zero. But often we care about deviations from the mean.

The **k-th central moment** of X is

$$\mathbb{E}[(X - \mu)^k],$$

where

$$\mu = \mathbb{E}[X].$$

The first central moment is

$$\mathbb{E}[X - \mu] = 0.$$

The second central moment is

$$\mathbb{E}[(X - \mu)^2] = \mathrm{Var}(X).$$

Thus, variance is the second central moment.

5.8 Skewness

The third central moment is

$$\mathbb{E}[(X - \mu)^3].$$

It measures asymmetry.

If the distribution has a longer right tail, the third central moment tends to be positive.

If the distribution has a longer left tail, the third central moment tends to be negative.

A standardized version is called **skewness**:

$$\mathrm{Skewness}(X) = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3},$$

where

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

In finance, skewness is important because asset returns are often not symmetric.

For example, a return distribution with negative skewness may have occasional large losses.

5.9 Kurtosis

The fourth central moment is

$$\mathbb{E}[(X-\mu)^4].$$

It measures tail heaviness and the tendency of a distribution to produce extreme values.

A standardized version is called **kurtosis**:

$$\text{Kurtosis}(X) = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4}.$$

The normal distribution has kurtosis 3. Therefore, people often define **excess kurtosis** as

$$\text{Excess Kurtosis} = \text{Kurtosis} - 3.$$

If a distribution has positive excess kurtosis, it has heavier tails than a normal distribution.

In finance, this matters because asset returns often have heavy tails. This means extreme gains or losses occur more often than a normal model would predict.

5.10 Moment Generating Function

The **moment generating function**, or MGF, of a random variable X is defined as

$$M_X(t) = \mathbb{E}[e^{tX}],$$

provided the expectation exists for t in some interval around 0.

The reason it is called the moment generating function is that derivatives of $M_X(t)$ at $t = 0$ give the moments of X.

Specifically,

$$M_X'(0) = \mathbb{E}[X],$$

$$M_X''(0) = \mathbb{E}[X^2],$$

and more generally,

$$M_X^{(k)}(0) = \mathbb{E}[X^k].$$

5.10.1 Why MGFs Are Useful

MGFs are useful because:

- They can help compute moments.
- They can identify distributions.

- They are useful for sums of independent random variables.

If X and Y are independent, then

$$M_{X+Y}(t) = M_X(t)M_Y(t).$$

This is one reason MGFs are useful in probability theory.

5.11 Covariance

Variance measures how much one random variable fluctuates around its own mean. Covariance measures how two random variables move together.

For random variables X and Y, covariance is defined as

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

An equivalent formula is

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

5.11.1 Interpretation

If

$$\mathrm{Cov}(X, Y) > 0,$$

then X and Y tend to move in the same direction.

If

$$\mathrm{Cov}(X, Y) < 0,$$

then X and Y tend to move in opposite directions.

If

$$\mathrm{Cov}(X, Y) = 0,$$

then X and Y are uncorrelated.

However, uncorrelated does not always mean independent.

5.12 Correlation

Covariance depends on the units of X and Y. To get a scale-free measure, we define correlation.

The correlation coefficient of X and Y is

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}.$$

It always satisfies

$$-1 \le \rho_{X,Y} \le 1.$$

If

$$\rho_{X,Y} = 1,$$

then X and Y are perfectly positively correlated.
If

$$\rho_{X,Y} = -1,$$

then X and Y are perfectly negatively correlated.
If

$$\rho_{X,Y} = 0,$$

then X and Y are uncorrelated.

In finance, correlation is important for portfolio construction because diversification depends on how asset returns move together.

5.13 Variance of a Sum

For two random variables X and Y,

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y).$$

More generally,

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i) + 2 \sum_{1\le i<j\le n} \operatorname{Cov}(X_i, X_j).$$

If $X_1, \dots, X_n$ are independent, then

$$\operatorname{Cov}(X_i, X_j) = 0 \qquad \text{for } i \neq j.$$

Therefore,

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i).$$

This result is very important for random walks. If

$$X_n = Y_1 + \cdots + Y_n,$$

where $Y_1, \dots, Y_n$ are iid with variance σ^2, then

$$\operatorname{Var}(X_n) = n\sigma^2.$$

Therefore, the standard deviation is

$$\mathrm{SD}(X_n) = \sqrt{n}\sigma.$$

This is the reason random walks typically fluctuate on the scale $\sqrt{n}$, not n, when they have zero drift.

5.14 Key Takeaways

- Expectation is the long-run average value of a random variable.
- Variance measures spread around the mean.
- Standard deviation is the square root of variance and is in the same units as the random variable.
- The k-th raw moment is $\mathbb{E}[X^k]$.
- The k-th central moment is $\mathbb{E}[(X-\mu)^k]$.
- Variance is the second central moment.
- Skewness measures asymmetry.
- Kurtosis measures tail heaviness.
- Covariance and correlation measure how two random variables move together.
- Linearity of expectation does not require independence.
- Variance of a sum only becomes a simple sum of variances when covariance terms vanish, for example under independence.

6 Stochastic Processes

6.1 Definition

A **stochastic process** is a collection of random variables indexed by time.

We usually write a stochastic process as

$$\{X_t : t \in T\},$$

where T is the time index set.

If time is discrete, we write

$$\{X_n : n = 0, 1, 2, \dots\}.$$

If time is continuous, we write

$$\{X_t : t \geq 0\}.$$

6.2 Examples

- X_n: price of a stock at day n.
- X_t: value of a portfolio at time t.
- N_t: number of trades up to time t.
- R_n: return of an asset on day n.
- S_n: position of a random walk after n steps.

6.3 Discrete-Time and Continuous-Time Processes

A stochastic process is **discrete-time** if observations are made at times

$$0, 1, 2, \ldots.$$

A stochastic process is **continuous-time** if observations are made at every time

$$t \geq 0.$$

In this week, we focus mainly on discrete-time processes.

7 Markov Chains

7.1 The Markov Property

A stochastic process $\{X_n\}_{n\geq 0}$ is called a **Markov chain** if the future depends on the past only through the present.

Formally,

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i).$$

This is called the **Markov property**.

In words:

Given the present, the future is independent of the past.

7.2 State Space

The set of possible values of a Markov chain is called the **state space**.

Examples:

- $S = \{0, 1\}$: default or no default.
- $S = \{0, 1, 2\}$: bear, neutral, bull market.
- $S = \{0, 1, 2, \ldots\}$: queue length.
- $S = \mathbb{Z}$: position of a random walk.

7.3 Transition Probabilities

The one-step transition probability from state i to state j is

$$p_{ij} = P(X_{n+1} = j \mid X_n = i).$$

If these probabilities do not depend on n, the Markov chain is called **time-homogeneous**.

7.4 Transition Probability Matrix

For a finite state space

$$S = \{1, 2, \ldots, m\},$$

the transition probabilities can be arranged in a matrix

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{pmatrix}.$$

This is called the **transition probability matrix**, or TPM.
For P to be valid:

$$p_{ij} \geq 0$$

for all i, j, and

$$\sum_{j=1}^{m} p_{ij} = 1$$

for every row i.

7.5 Example: Two-State Markov Chain

Suppose a market can be in two states:

$$0 = \text{Down}, \qquad 1 = \text{Up}.$$

Let

$$P = \begin{pmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{pmatrix}.$$

This means:

$$P(X_{n+1} = 0 \mid X_n = 0) = 0.6,$$

$$P(X_{n+1} = 1 \mid X_n = 0) = 0.4,$$

$$P(X_{n+1} = 0 \mid X_n = 1) = 0.3,$$

$$P(X_{n+1} = 1 \mid X_n = 1) = 0.7.$$

7.6 Multi-Step Transition Probabilities

The n-step transition probability is

$$p_{ij}^{(n)} = P(X_n = j \mid X_0 = i).$$

For a time-homogeneous Markov chain,

$$p_{ij}^{(n)} = (P^n)_{ij}.$$

Thus, two-step transition probabilities are found from P^2, three-step transition probabilities from P^3, and so on.

7.7 Distribution Evolution

Suppose the initial distribution is a row vector

$$\pi^{(0)}.$$

Then after one step,

$$\pi^{(1)} = \pi^{(0)} P.$$

After n steps,

$$\pi^{(n)} = \pi^{(0)} P^n.$$

8 Class Properties of Markov Chains

In a Markov chain, the transition probability matrix tells us how the chain moves from one state to another in one step. However, many important properties of a Markov chain are not about one-step movement alone. They are about which states can eventually reach which other states, whether the chain can get trapped, whether it returns regularly, and whether long-run behaviour exists.

These are called **class properties**.

8.1 Multi-Step Reachability

The one-step transition probabilities are entries of P. The two-step transition probabilities are entries of P^2. More generally, the n-step transition probabilities are entries of P^n.

We write

$$p_{ij}^{(n)} = P(X_n = j \mid X_0 = i).$$

For a time-homogeneous Markov chain,

$$p_{ij}^{(n)} = (P^n)_{ij}.$$

Thus, to understand whether state j can be reached from state i, we look at whether

$$p_{ij}^{(n)} > 0$$

for some $n \geq 0$.

8.2 Accessibility

A state j is said to be **accessible** from state i if it is possible to reach j starting from i in some number of steps.

We write

$$i \to j$$

if j is accessible from i.

Mathematically,

$$i \to j$$

if there exists some $n \geq 0$ such that

$$p_{ij}^{(n)} > 0.$$

Equivalently,

$$(P^n)_{ij} > 0$$

for some $n \geq 0$.

8.3 Communication

Two states i and j are said to **communicate** if each is accessible from the other.

We write

$$i \leftrightarrow j$$

if

$$i \to j$$

and

$$j \to i.$$

Communication means that it is possible to go from i to j, and also possible to return from j to i.

8.4 Communicating Classes

Communication divides the state space into groups called **communicating classes**.

A communicating class is a set of states such that:

- every state in the set communicates with every other state in the set;
- no larger set with this property contains it.

In simple terms, a communicating class is a group of states that can all reach each other.

8.5 Example: Communicating Classes

Consider

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.4 & 0.6 & 0 & 0 \\ 0 & 0 & 0.3 & 0.7 \\ 0 & 0 & 0.2 & 0.8 \end{pmatrix}.$$

States 1 and 2 communicate with each other. States 3 and 4 communicate with each other. But there is no way to move between the pair $\{1,2\}$ and the pair $\{3,4\}$.

Therefore, the communicating classes are

$$\{1,2\}$$

and

$$\{3,4\}.$$

8.6 Irreducibility

A Markov chain is called **irreducible** if all states communicate with each other.

That is, for every pair of states $i, j \in S$,

$$i \leftrightarrow j.$$

Equivalently, the entire state space consists of one communicating class.

8.6.1 Interpretation

Irreducibility means that the chain is not split into separate disconnected parts. Starting from any state, it is possible to eventually reach any other state.

8.6.2 Example of an Irreducible Chain

Consider

$$P = \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix}.$$

From state 1, we can reach state 2 with probability 0.8. From state 2, we can reach state 1 with probability 0.5. Therefore,

$$1 \leftrightarrow 2.$$

Since there are only two states, the chain is irreducible.

8.6.3 Example of a Reducible Chain

Consider

$$P = \begin{pmatrix} 1 & 0 \\ 0.4 & 0.6 \end{pmatrix}.$$

From state 2, it is possible to reach state 1. But from state 1, the chain stays in state 1 forever. Therefore, state 2 is not accessible from state 1.

So the chain is not irreducible. It is reducible.

8.7 Closed Classes

A communicating class C is called **closed** if, once the chain enters C, it cannot leave C.

Mathematically, C is closed if

$$p_{ij} = 0$$

for all

$$i \in C, \qquad j \notin C.$$

In other words, there is no transition from a state inside C to a state outside C.

8.7.1 Example

Consider

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.4 & 0.6 & 0 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}.$$

The set

$$C = \{1, 2\}$$

is closed because from states 1 and 2, the chain never moves to state 3.

However, the set

$$\{3\}$$

is not closed because from state 3, the chain can move to states 1 and 2.

8.8 Absorbing States

A state i is called **absorbing** if, once the chain enters state i, it stays there forever.

Mathematically, state i is absorbing if

$$p_{ii} = 1.$$

Equivalently,

$$P(X_{n+1} = i \mid X_n = i) = 1.$$

An absorbing state is a closed communicating class containing exactly one state.

8.8.1 Example

Consider

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0.3 & 0.4 & 0.3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Here,

$$p_{11} = 1$$

and

$$p_{33} = 1.$$

Therefore, states 1 and 3 are absorbing.

8.9 Transient and Recurrent States

A state is called **recurrent** if, starting from that state, the chain returns to it with probability 1.

A state is called **transient** if there is a positive probability that the chain never returns to it.

Let

$$T_i = \inf\{n \geq 1 : X_n = i\}$$

be the first return time to state i. Then state i is recurrent if

$$P(T_i < \infty \mid X_0 = i) = 1.$$

It is transient if

$$P(T_i < \infty \mid X_0 = i) < 1.$$

8.9.1 Intuition

A recurrent state is one that the chain is guaranteed to revisit eventually.

A transient state is one that the chain may leave and never come back to.

8.9.2 Important Fact for Finite Chains

In a finite Markov chain:

- every state in a closed communicating class is recurrent;
- states that can lead into a closed class but cannot be reached back from it are transient.

8.10 Periodicity

The **period** of a state i is the greatest common divisor of all times at which it is possible to return to i.

Mathematically, the period of state i is

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

If

$$d(i) = 1,$$

then state i is called **aperiodic**.

If

$$d(i) > 1,$$

then state i is called **periodic** with period $d(i)$.

8.10.1 Example: Period 2

Consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Starting from state 1, the chain moves to state 2, then back to state 1, then to state 2, and so on.

The chain can return to state 1 only at times

$$2, 4, 6, \dots$$

Therefore,

$$d(1) = \gcd\{2, 4, 6, \dots\} = 2.$$

So state 1 has period 2.

8.10.2 Example: Aperiodic State

Consider

$$P = \begin{pmatrix} 0.5 & 0.5 \\ 0.4 & 0.6 \end{pmatrix}.$$

Since

$$p_{11} > 0,$$

it is possible to return to state 1 in one step. Therefore,

$$1 \in \{n \geq 1 : p_{11}^{(n)} > 0\}.$$

Hence,

$$d(1) = 1.$$

So state 1 is aperiodic.

8.11 Periodicity as a Class Property

If two states communicate, then they have the same period.

Therefore, periodicity is a **class property**. This means that all states in the same communicating class have the same period.

In particular, if a Markov chain is irreducible, then all states have the same period. Therefore, we can speak of the period of the chain.

8.12 Positive Recurrence

A recurrent state i is called **positive recurrent** if the expected return time to i is finite.

Let

$$T_i = \inf\{n \geq 1 : X_n = i\}$$

be the first return time to i. State i is positive recurrent if

$$\mathbb{E}[T_i \mid X_0 = i] < \infty.$$

For finite irreducible Markov chains, every state is positive recurrent.

This fact is important because positive recurrence is closely related to the existence of stationary distributions.

8.13 Null Recurrence

A recurrent state is called **null recurrent** if the chain returns to the state with probability 1, but the expected return time is infinite.

That is, state i is null recurrent if

$$P(T_i < \infty \mid X_0 = i) = 1,$$

but

$$\mathbb{E}[T_i \mid X_0 = i] = \infty.$$

Null recurrence does not occur in finite irreducible Markov chains, but it can occur in infinite state spaces.

For example, the simple symmetric random walk on $\mathbb{Z}$ is recurrent but has infinite expected return time.

8.14 Class Properties

A property is called a **class property** if whenever one state in a communicating class has the property, then every state in that communicating class also has the property.

In other words, a class property is shared by all states that communicate with each other.

Important class properties include:

- recurrence;
- transience;
- positive recurrence;
- null recurrence;
- periodicity;

- aperiodicity.

This means that if $i \leftrightarrow j$, then:

$$i \text{ is recurrent} \iff j \text{ is recurrent},$$

$$i \text{ is transient} \iff j \text{ is transient},$$

and

$$d(i) = d(j).$$

8.15 Ergodic Markov Chains

A finite Markov chain is often called **ergodic** if it is:

- irreducible;
- aperiodic.

Some texts also include positive recurrence in the definition. For finite chains, irreducibility already implies positive recurrence.

Ergodicity is important because it gives strong long-run behaviour.

If a finite Markov chain is irreducible and aperiodic, then it has a unique stationary distribution π, and

$$P^n(i,j) \to \pi_j$$

as

$$n \to \infty.$$

This means that the long-run probability of being in state j does not depend on the starting state.

8.16 How to Analyze Class Properties from a TPM

Given a transition probability matrix P, follow these steps:

1. Draw the directed graph of the chain:

$$i \to j \quad \text{if} \quad p_{ij} > 0.$$

2. Find which states can reach which other states.
3. Group states into communicating classes.
4. Check which classes are closed.

5. Identify absorbing states, if any.

6. Determine whether the chain is irreducible.

7. Check periodicity by finding possible return times to states.

8. Use the class structure to reason about recurrence, transience, and long-run behaviour.

8.17 Example: Full Class Analysis

Consider the transition matrix

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.4 & 0.6 & 0 & 0 \\ 0.2 & 0 & 0.3 & 0.5 \\ 0 & 0 & 0.7 & 0.3 \end{pmatrix}.$$

The state space is

$$S = \{1, 2, 3, 4\}.$$

From states 1 and 2, the chain stays within $\{1, 2\}$. Also, states 1 and 2 can reach each other. Therefore,

$$\{1, 2\}$$

is a communicating class.

From states 3 and 4, the chain can move between 3 and 4. Thus,

$$\{3, 4\}$$

is also a communicating class. However, from state 3, the chain can move to state 1. Once it enters $\{1, 2\}$, it cannot return to $\{3, 4\}$.

Therefore:

$$\{1, 2\}$$

is a closed communicating class, while

$$\{3, 4\}$$

is not closed.

Since the chain can move from $\{3, 4\}$ into $\{1, 2\}$, but not back, states 3 and 4 are transient.

States 1 and 2 are recurrent because they form a closed class in a finite Markov chain.

The chain is not irreducible because not all states communicate with each other.

Also, since

$$p_{11} > 0$$

and

$$p_{22} > 0,$$

states 1 and 2 are aperiodic. Similarly,

$$p_{33} > 0$$

and

$$p_{44} > 0,$$

so states 3 and 4 are also aperiodic.

8.18 Key Takeaways

- Accessibility means one state can be reached from another.
- Communication means two states can reach each other.
- Communicating classes are groups of mutually reachable states.
- A chain is irreducible if the whole state space is one communicating class.
- A closed class cannot be left once entered.
- An absorbing state is a one-state closed class.
- Recurrence and transience describe whether states are revisited.
- Periodicity describes the possible return times to a state.
- Recurrence, transience, positive recurrence, null recurrence, and periodicity are class properties.
- In finite irreducible Markov chains, all states are positive recurrent.
- Finite irreducible and aperiodic chains have strong long-run convergence to a unique stationary distribution.

9 Stationary Distributions

9.1 Definition

A probability vector π is called a **stationary distribution** of a Markov chain with transition matrix P if

$$\pi P = \pi.$$

Also,

$$\sum_i \pi_i = 1, \qquad \pi_i \geq 0.$$

If the chain starts with distribution π, then after one step it still has distribution π.

9.2 Interpretation

A stationary distribution describes the long-run proportion of time the chain spends in each state, under suitable conditions.

For example, if

$$\pi = (0.2, 0.5, 0.3),$$

then in the long run, the chain spends approximately:

- 20% of the time in state 1,
- 50% of the time in state 2,
- 30% of the time in state 3.

9.3 Example

Let

$$P = \begin{pmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{pmatrix}.$$

Let

$$\pi = (a, b).$$

We need

$$\pi P = \pi$$

and

$$a + b = 1.$$

Now,

$$(a, b) \begin{pmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{pmatrix} = (a, b).$$

This gives

$$0.6a + 0.3b = a,$$

$$0.4a + 0.7b = b.$$

Using the first equation,

$$0.3b = 0.4a.$$

Thus,

$$b = \frac{4}{3}a.$$

Using $a + b = 1$,

$$a + \frac{4}{3}a = 1.$$

So

$$\frac{7}{3}a = 1,$$

and hence

$$a = \frac{3}{7}.$$

Therefore,

$$b = \frac{4}{7}.$$

The stationary distribution is

$$\pi = \left(\frac{3}{7}, \frac{4}{7}\right).$$

9.4 Stationary Distribution and Class Properties

Class properties help us understand when stationary distributions exist and when they are unique.

For a finite Markov chain:

- If the chain is irreducible, then a unique stationary distribution exists.
- If the chain is irreducible and aperiodic, then the distribution of X_n converges to the stationary distribution.
- If the chain is reducible, there may be multiple stationary distributions.
- Closed communicating classes play a major role in determining long-run behaviour.

10 Random Walks

10.1 Motivation

Random walks are among the most important examples of stochastic processes. They are simple to define, but they appear in many areas of probability, statistics, physics, finance, and computer science.

The basic idea is this:

$$\text{current position} = \text{previous position} + \text{random step}.$$

In finance, random walks are useful because they provide a simple way to model uncertain price movements. A price may go up or down randomly from one time period to the next. Although real financial markets are much more complicated, random walks are a natural first model for randomness evolving over time.

Random walks are also important because Brownian motion can be obtained as a scaling limit of random walks. Brownian motion is one of the central stochastic processes used in continuous-time finance.

10.2 General Definition of a Random Walk

Let

$$Y_1, Y_2, \dots$$

be independent and identically distributed random variables. These are called the **increments** or **steps** of the walk.

Define

$$X_0 = 0,$$

and for $n \geq 1$,

$$X_n = Y_1 + Y_2 + \cdots + Y_n.$$

Equivalently,

$$X_n = X_{n-1} + Y_n.$$

The stochastic process

$$\{X_n : n \geq 0\}$$

is called a **random walk**.

The value X_n is the position of the walk after n steps.

10.3 Interpretation of the Increments

The increments Y_i determine the behaviour of the random walk.

- If $\mathbb{E}[Y_i] > 0$, the random walk has a positive drift.
- If $\mathbb{E}[Y_i] < 0$, the random walk has a negative drift.
- If $\mathbb{E}[Y_i] = 0$, the random walk has zero drift.

The variance of the increments determines how much the walk fluctuates around its mean path.

$$\mathrm{Var}(X_n) = n\,\mathrm{Var}(Y_1)$$

when the increments are independent.

Thus, even when the mean is zero, the random walk spreads out over time.

10.4 Simple Symmetric Random Walk on $\mathbb{Z}$

The most basic random walk is the **simple symmetric random walk** on the integers.

Here the state space is

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

At each step, the walk moves either one unit to the right or one unit to the left with equal probability.

Define

$$Y_i = \begin{cases} 1, & \text{with probability } \frac{1}{2}, \\ -1, & \text{with probability } \frac{1}{2}. \end{cases}$$

Then

$$X_n = Y_1 + \cdots + Y_n.$$

This is called the **one-dimensional simple symmetric random walk**, or **1D SSRW**.

10.4.1 Transition Probabilities

If the current position is $i \in \mathbb{Z}$, then

$$P(X_{n+1} = i + 1 \mid X_n = i) = \frac{1}{2},$$

and

$$P(X_{n+1} = i - 1 \mid X_n = i) = \frac{1}{2}.$$

All other one-step transition probabilities are zero.

So,

$$p_{ij} = \begin{cases} \frac{1}{2}, & j = i + 1, \\ \frac{1}{2}, & j = i - 1, \\ 0, & \text{otherwise.} \end{cases}$$

10.4.2 Example Path

Suppose the first six increments are

$$Y_1 = 1, \quad Y_2 = -1, \quad Y_3 = 1, \quad Y_4 = 1, \quad Y_5 = -1, \quad Y_6 = 1.$$

Then the positions are:

$$X_0 = 0,$$

$$X_1 = 1,$$

$$X_2 = 1 + (-1) = 0,$$

$$X_3 = 1,$$

$$X_4 = 2,$$

$$X_5 = 1,$$

$$X_6 = 2.$$

So the path is

$$0, 1, 0, 1, 2, 1, 2.$$

10.4.3 Expectation

For each step,

$$\mathbb{E}[Y_i] = 1 \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0.$$

Using linearity of expectation,

$$\mathbb{E}[X_n] = \mathbb{E}[Y_1 + \cdots + Y_n] = \sum_{i=1}^{n} \mathbb{E}[Y_i] = 0.$$

Thus, the expected position after n steps is

$$\mathbb{E}[X_n] = 0.$$

This does not mean the walk stays near zero. It means the average position over many independent walks is zero.

10.4.4 Variance

Since

$$Y_i^2 = 1$$

always, we have

$$\mathbb{E}[Y_i^2] = 1.$$

Also,

$$\mathbb{E}[Y_i] = 0.$$

Therefore,

$$\mathrm{Var}(Y_i) = \mathbb{E}[Y_i^2] - (\mathbb{E}[Y_i])^2 = 1 - 0 = 1.$$

Since the increments are independent,

$$\mathrm{Var}(X_n) = \mathrm{Var}(Y_1 + \cdots + Y_n) = \sum_{i=1}^{n} \mathrm{Var}(Y_i) = n.$$

Therefore,

$$\mathrm{SD}(X_n) = \sqrt{n}.$$

This is one of the most important facts about random walks:

$$\text{typical size of displacement after } n \text{ steps is of order } \sqrt{n}.$$

So even though the walk takes n steps, it does not usually end up distance n away from the origin. Because positive and negative steps cancel each other, the typical displacement is much smaller, of order $\sqrt{n}$.

10.4.5 Distribution of X_n

In the 1D Simple Symmetric Random Walk (SSRW), after n steps, suppose the walk has taken k upward/rightward steps and $n-k$ downward/leftward steps.

Then

$$X_n = k - (n-k) = 2k - n.$$

Therefore,

$$X_n = 2k - n.$$

This means X_n can only take values with the same parity as n. For example:

- after $n = 4$ steps, possible positions are $-4, -2, 0, 2, 4$;
- after $n = 5$ steps, possible positions are $-5, -3, -1, 1, 3, 5$.

The probability that $X_n = 2k - n$ is

$$P(X_n = 2k - n) = \binom{n}{k} \left(\frac{1}{2}\right)^n.$$

This follows because choosing which k of the n steps are $+1$ determines the path.

10.5 Biased Random Walk on $\mathbb{Z}$

A biased random walk is similar, except that the probabilities of moving right and left are not equal.

Let

$$Y_i = \begin{cases} 1, & \text{with probability } p, \\ -1, & \text{with probability } 1-p. \end{cases}$$

Then

$$X_n = Y_1 + \cdots + Y_n.$$

The expected step is

$$\mathbb{E}[Y_i] = 1 \cdot p + (-1)(1-p) = 2p - 1.$$

Therefore,

$$\mathbb{E}[X_n] = n(2p-1).$$

If $p > \frac{1}{2}$, the walk drifts upward/rightward.
If $p < \frac{1}{2}$, the walk drifts downward/leftward.
If $p = \frac{1}{2}$, the walk is symmetric and has zero drift.
The variance of one step is

$$\mathrm{Var}(Y_i) = 1 - (2p-1)^2 = 4p(1-p).$$

Therefore,

$$\mathrm{Var}(X_n) = 4np(1-p).$$

10.6 Random Walks in Higher Dimensions

Random walks can also be defined in higher-dimensional spaces such as

$$\mathbb{Z}^2$$

and

$$\mathbb{Z}^3.$$

In higher dimensions, the position is a vector rather than a single number.
For example, in two dimensions,

$$X_n = (X_n^{(1)}, X_n^{(2)}) \in \mathbb{Z}^2.$$

In three dimensions,

$$X_n = (X_n^{(1)}, X_n^{(2)}, X_n^{(3)}) \in \mathbb{Z}^3.$$

10.7 Two-Dimensional Simple Symmetric Random Walk

The **two-dimensional simple symmetric random walk**, or **2D SSRW**, takes place on the integer lattice

$$\mathbb{Z}^2.$$

The walk starts at

$$X_0 = (0,0).$$

At each step, it moves to one of its four nearest neighbours with equal probability.
The possible steps are

$$(1,0), \quad (-1,0), \quad (0,1), \quad (0,-1).$$

Each occurs with probability

$$\frac{1}{4}.$$

Thus,

$$Y_i = \begin{cases} (1,0), & \text{with probability } \frac{1}{4}, \\ (-1,0), & \text{with probability } \frac{1}{4}, \\ (0,1), & \text{with probability } \frac{1}{4}, \\ (0,-1), & \text{with probability } \frac{1}{4}. \end{cases}$$

The position after n steps is

$$X_n = Y_1 + \cdots + Y_n.$$

10.7.1 Transition Probabilities

If the current position is

$$(x, y) \in \mathbb{Z}^2,$$

then the next position is one of

$$(x+1, y), \quad (x-1, y), \quad (x, y+1), \quad (x, y-1),$$

each with probability 1/4.

So,

$$P(X_{n+1} = (x+1, y) \mid X_n = (x, y)) = \frac{1}{4},$$

$$P(X_{n+1} = (x-1, y) \mid X_n = (x, y)) = \frac{1}{4},$$

$$P(X_{n+1} = (x, y+1) \mid X_n = (x, y)) = \frac{1}{4},$$

and

$$P(X_{n+1} = (x, y-1) \mid X_n = (x, y)) = \frac{1}{4}.$$

10.7.2 Example Path

Suppose the first five steps are

$$(1,0), \quad (0,1), \quad (-1,0), \quad (0,1), \quad (1,0).$$

Then the positions are

$$X_0 = (0,0),$$

$$X_1 = (1,0),$$

$$X_2 = (1, 1),$$

$$X_3 = (0, 1),$$

$$X_4 = (0, 2),$$

$$X_5 = (1, 2).$$

So the walk moves on the two-dimensional grid.

10.7.3 Mean and Spread

For one step $Y_i = (Y_i^{(1)}, Y_i^{(2)})$,

$$\mathbb{E}[Y_i] = (0, 0).$$

Therefore,

$$\mathbb{E}[X_n] = (0, 0).$$

Although the expected position is the origin, the walk spreads out over time.
The squared distance from the origin is

$$\|X_n\|^2 = \left(X_n^{(1)}\right)^2 + \left(X_n^{(2)}\right)^2.$$

For the 2D SSRW, one can show that

$$\mathbb{E}[\|X_n\|^2] = n.$$

Thus, the typical distance from the origin is still of order

$$\sqrt{n}.$$

10.8 Three-Dimensional Simple Symmetric Random Walk

The **three-dimensional simple symmetric random walk**, or **3D SSRW**, takes place on

$$\mathbb{Z}^3.$$

The walk starts at

$$X_0 = (0, 0, 0).$$

At each step, it moves to one of its six nearest neighbours with equal probability.
The possible steps are

$$(1,0,0), \quad (-1,0,0),$$

$$(0,1,0), \quad (0,-1,0),$$

$$(0,0,1), \quad (0,0,-1).$$

Each occurs with probability

$$\frac{1}{6}.$$

Thus, the walk moves one unit in either the positive or negative direction along one of the three coordinate axes.

10.8.1 Transition Probabilities

If the current position is

$$(x,y,z) \in \mathbb{Z}^3,$$

then the next position is one of

$$(x+1,y,z), \quad (x-1,y,z),$$

$$(x,y+1,z), \quad (x,y-1,z),$$

$$(x,y,z+1), \quad (x,y,z-1),$$

each with probability 1/6.

10.8.2 Example Path

Suppose the first four steps are

$$(1,0,0), \quad (0,1,0), \quad (0,0,-1), \quad (-1,0,0).$$

Then the positions are

$$X_0 = (0,0,0),$$

$$X_1 = (1,0,0),$$

$$X_2 = (1,1,0),$$

$$X_3 = (1, 1, -1),$$

$$X_4 = (0, 1, -1).$$

10.8.3 Mean and Spread

For one step Y_i,

$$\mathbb{E}[Y_i] = (0, 0, 0).$$

Therefore,

$$\mathbb{E}[X_n] = (0, 0, 0).$$

Again, the expected position is the origin, but the walk spreads out.

For the 3D SSRW,

$$\mathbb{E}[\|X_n\|^2] = n.$$

So the typical distance from the origin is of order

$$\sqrt{n}.$$

10.9 General d-Dimensional Simple Symmetric Random Walk

The d-dimensional simple symmetric random walk takes place on

$$\mathbb{Z}^d.$$

The walk starts at the origin

$$X_0 = (0, 0, \ldots, 0).$$

At each step, it chooses one of the d coordinate directions and moves either $+1$ or -1 in that coordinate.

There are $2d$ possible moves:

$$\pm e_1, \pm e_2, \ldots, \pm e_d,$$

where e_i is the i-th standard basis vector.

Each move has probability

$$\frac{1}{2d}.$$

Thus,

$$P(Y_n = e_i) = \frac{1}{2d}, \qquad P(Y_n = -e_i) = \frac{1}{2d},$$

for

$$i = 1, 2, \ldots, d.$$

The position is

$$X_n = Y_1 + \cdots + Y_n.$$

For the d-dimensional SSRW,

$$\mathbb{E}[Y_i] = 0,$$

and hence

$$\mathbb{E}[X_n] = 0.$$

Also,

$$\mathbb{E}[\|X_n\|^2] = n.$$

Therefore, the typical distance from the origin is of order

$$\sqrt{n}.$$

10.10 Recurrence and Transience of Random Walks

One of the most interesting questions about random walks is:

Will the walk return to its starting point?

For simple symmetric random walks, the answer depends strongly on the dimension.

- In one dimension, the simple symmetric random walk returns to the origin with probability 1.
- In two dimensions, the simple symmetric random walk also returns to the origin with probability 1.
- In three dimensions, the simple symmetric random walk returns to the origin with probability strictly less than 1.

Thus:

1D and 2D SSRW are recurrent,

while

3D SSRW is transient.

This is a deep and important result. Intuitively, in one and two dimensions, the walk does not have enough space to escape forever. In three or more dimensions, there is enough space for the walk to drift away and possibly never return.

10.11 Random Walks as Markov Chains

Random walks are Markov chains because the next position depends only on the current position and the next random increment.

For example, in the 1D SSRW,

$$P(X_{n+1} = i + 1 \mid X_n = i) = \frac{1}{2},$$

and

$$P(X_{n+1} = i - 1 \mid X_n = i) = \frac{1}{2}.$$

The past positions

$$X_0, X_1, \ldots, X_{n-1}$$

do not matter once X_n is known.

Similarly, in the 2D SSRW, if the current position is (x, y), the next position is one of its four neighbours, each with probability $1/4$, regardless of how the walk reached (x, y).

Thus, random walks are among the most natural examples of Markov chains.

10.12 Random Walks and Finance

Random walks are important in finance because they provide a simple model for uncertain price movements.

A basic additive price model is

$$S_{n+1} = S_n + Y_{n+1},$$

where:

- S_n is the asset price at time n,
- Y_{n+1} is the random price change from time n to time $n + 1$.

A multiplicative return model is

$$S_{n+1} = S_n(1 + R_{n+1}),$$

where R_{n+1} is the random return.

Often, it is more convenient to model log-prices. If

$$Z_n = \log S_n,$$

then a simple model is

$$Z_{n+1} = Z_n + Y_{n+1}.$$

This means the log-price follows a random walk.

Random walks are not perfect models of real markets, but they are foundational. More advanced models, such as Brownian motion, geometric Brownian motion, and stochastic volatility models, build on the same idea of random increments over time.

10.13 Connection to Brownian Motion

A simple symmetric random walk has

$$\mathbb{E}[X_n] = 0$$

and

$$\mathrm{Var}(X_n) = n.$$

To obtain a non-trivial continuous-time limit, we scale time and space together.

A rough idea is:

$$B_t^{(n)} = \frac{X_{\lfloor nt \rfloor}}{\sqrt{n}}.$$

As

$$n \to \infty,$$

this scaled random walk converges in distribution to Brownian motion.

This result is formalized by Donsker's theorem.

Brownian motion is central in continuous-time finance. In the Black–Scholes model, for example, stock prices are driven by Brownian motion.

10.14 Key Takeaways

- A random walk is formed by adding random increments.
- The position after n steps is $X_n = Y_1 + \cdots + Y_n$.
- A simple symmetric random walk has equally likely moves in opposite directions.
- In 1D, the SSRW moves left or right on $\mathbb{Z}$.
- In 2D, the SSRW moves to one of four nearest neighbours on $\mathbb{Z}^2$.
- In 3D, the SSRW moves to one of six nearest neighbours on $\mathbb{Z}^3$.

- For zero-drift random walks, the expected position is zero.
- The variance grows linearly in n.
- The typical displacement is of order $\sqrt{n}$.
- 1D and 2D SSRW are recurrent, while 3D SSRW is transient.
- Random walks are Markov chains.
- Brownian motion can be viewed as a scaling limit of random walks.
- Random walks are foundational models for stochastic finance.

11 Simulation in Python

Simulation helps us see stochastic processes in action. It is especially useful when exact calculations are difficult.

11.1 Simulating a Simple Symmetric Random Walk

```python
import numpy as np
import matplotlib.pyplot as plt

n = 1000

steps = np.random.choice([1, -1], size=n, p=[0.5, 0.5])
path = np.cumsum(steps)

plt.figure(figsize=(10, 5))
plt.plot(path)
plt.xlabel("Step")
plt.ylabel("Position")
plt.title("Simple Symmetric Random Walk")
plt.grid(True)
plt.show()
```

11.2 Simulating Many Random Walks

```python
import numpy as np
import matplotlib.pyplot as plt

num_paths = 10000
n = 1000

steps = np.random.choice([1, -1], size=(num_paths, n), p=[0.5, 0.5])
final_positions = steps.sum(axis=1)

```

```python
print("Empirical mean:", np.mean(final_positions))
print("Empirical variance:", np.var(final_positions))

plt.figure(figsize=(10, 5))
plt.hist(final_positions, bins=50)
plt.xlabel("Final Position")
plt.ylabel("Frequency")
plt.title("Histogram of Final Positions")
plt.grid(True)
plt.show()
```

For a simple symmetric random walk,

$$\mathbb{E}[X_n] = 0$$

and

$$\mathrm{Var}(X_n) = n.$$

So for $n = 1000$, the empirical mean should be close to 0, and the empirical variance should be close to 1000.

11.3 Simulating a Biased Random Walk

```python
import numpy as np
import matplotlib.pyplot as plt

n = 1000
p = 0.55

steps = np.random.choice([1, -1], size=n, p=[p, 1-p])
path = np.cumsum(steps)

plt.figure(figsize=(10, 5))
plt.plot(path)
plt.xlabel("Step")
plt.ylabel("Position")
plt.title("Biased Random Walk")
plt.grid(True)
plt.show()
```

If $p = 0.55$, then

$$\mathbb{E}[Y_i] = 2p - 1 = 0.10.$$

Therefore,

$$\mathbb{E}[X_n] = 0.10n.$$

The random walk has upward drift.

11.4 Simulating a Finite Markov Chain

Suppose the transition matrix is

$$P = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}.$$

The states are:

$$0 = \text{Bear}, \qquad 1 = \text{Neutral}, \qquad 2 = \text{Bull}.$$

```
import numpy as np

P = np.array([
    [0.6, 0.3, 0.1],
    [0.2, 0.6, 0.2],
    [0.1, 0.3, 0.6]
])

states = [0, 1, 2]

n = 100
path = np.zeros(n, dtype=int)

path[0] = 1

 for t in range(1, n):
    current_state = path[t-1]
    path[t] = np.random.choice(states, p=P[current_state])

print(path)
```

Note: In Python, indentation matters. The loop should be written without an extra leading space before `for`:

```
for t in range(1, n):
    current_state = path[t-1]
    path[t] = np.random.choice(states, p=P[current_state])
```

11.5 Estimating Long-Run Frequencies

```
import numpy as np

P = np.array([
    [0.6, 0.3, 0.1],
```

```
    [0.2, 0.6, 0.2],
    [0.1, 0.3, 0.6]
])

states = [0, 1, 2]
n = 100000

path = np.zeros(n, dtype=int)
path[0] = 1

for t in range(1, n):
    current_state = path[t-1]
    path[t] = np.random.choice(states, p=P[current_state])

frequencies = np.bincount(path, minlength=3) / n

print("Long-run frequencies:", frequencies)
```

For an irreducible and aperiodic finite Markov chain, these long-run frequencies should be close to the stationary distribution.

12 Connection to Brownian Motion

Brownian motion is a continuous-time stochastic process that can be viewed as a scaling limit of random walks.

A simple symmetric random walk has

$$\mathbb{E}[X_n] = 0$$

and

$$\operatorname{Var}(X_n) = n.$$

If we scale space by $\sqrt{n}$, then the random walk begins to resemble Brownian motion.

Very roughly,

$$\frac{X_{\lfloor nt \rfloor}}{\sqrt{n}}$$

converges in distribution to Brownian motion as

$$n \to \infty.$$

This idea is formalized by Donsker's theorem.

Brownian motion is central in mathematical finance because it is used to model continuous-time random fluctuations in asset prices.

13 Week 1 Summary

The main ideas from Week 1 are:

- Probability gives a mathematical language for uncertainty.
- Random variables assign numerical values to random outcomes.
- Probability distributions describe how random variables behave.
- Expectation measures long-run average value.
- Variance and standard deviation measure spread.
- Moments describe the shape of a distribution.
- Skewness measures asymmetry.
- Kurtosis measures tail heaviness.
- Stochastic processes describe random quantities evolving over time.
- Markov chains are stochastic processes where the future depends on the past only through the present.
- Transition probability matrices describe one-step movement between states.
- Multi-step transition probabilities are obtained from powers of the transition matrix.
- Class properties describe the structure and long-run behaviour of Markov chains.
- Communicating classes group states that can reach each other.
- Closed classes cannot be left once entered.
- Irreducibility means the whole state space is one communicating class.
- Recurrence and transience describe whether states are revisited.
- Periodicity describes possible return times.
- Stationary distributions describe long-run behaviour.
- Random walks are fundamental examples of Markov chains.
- For zero-drift random walks, variance grows linearly in n, so typical fluctuations are of order $\sqrt{n}$.
- Simulations help connect theory with observed stochastic behaviour.

14 Suggested External References

The following references are optional but useful for deeper understanding:

- Harvard Stat 110: Probability.
- Harvard Stat 110 YouTube lectures.
- MIT OCW 6.436J Fundamentals of Probability.
- MIT OCW 6.041 Probabilistic Systems Analysis and Applied Probability.
- Grinstead and Snell, *Introduction to Probability*.
- MIT OCW Topics in Mathematics with Applications in Finance.